Frequency-Domain Descriptions for Discrete-Time Linear Time-Invariant Systems

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Key Concepts

1) Frequency response $H(e^{j\omega})$ is the Fourier transform of the impulse response.

    a) The magnitude response is $|H(e^{j\omega})|$.

    b) The phase response is $\arg\{H(e^{j\omega})\}$.

    c) If the input is a complex sinusoid, the output is has its amplitude scaled by $|H(e^{j\omega})|$ and its phase shifted by $\arg\{H(e^{j\omega})\}$.

2) If a system is described by the difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

then the frequency response is a ratio of polynomials in $e^{-j\omega}$

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}}$$

3) The frequency response of a system described by a difference equation is equal to the system function evaluated at $z = e^{j\omega}$.

4) The impulse response of a system described by a difference equation

may be obtained as the inverse discrete-time Fourier transform of the frequency response.

a) Find the roots d_ℓ of the characteristic equation

$$a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N} = 0 = a_0 \prod_{\ell=1}^{N} (1 - d_\ell z^{-1})$$

b) If the roots d_ℓ are distinct and $M < N$, then the partial fraction expansion is

$$H(e^{j\omega}) = \frac{A_1}{1 - d_1 e^{-j\omega}} + \frac{A_2}{1 - d_2 e^{-j\omega}} + \cdots + \frac{A_N}{1 - d_N e^{-j\omega}}$$

c) Solve for the A_i

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d) Take the inverse discrete-time Fourier transform of each term

$$h(t) = A_1 d_1^n u[n] + A_2 d_2^n u[n] + \ldots + A_N d_N^n u[n]$$

e) This requires the roots $|d_\ell| < 1$, which is also the condition for the system to be stable.

f) If the roots are repeated, a similar, but more complex partial fraction expansion is used.

g) If $M \geq N$, then long division must be used prior to partial fraction expansion.

h) If a_k, b_k are real, then the roots d_ℓ and the weights A_ℓ occur in complex conjugate pairs.

Video Screenshots

Linear time-invariant system descriptions

1) Impulse Response



$$y[n] = h[n] * x[n]$$

2) Difference Equation

$$y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = b_0 x[n] + \cdots + b_M x[n-M]$$

Frequency Response $H(e^{j\omega})$

$$y[n] = h[n] * x[n] \quad \overset{DTFT}{\longleftrightarrow} \quad Y(e^{j\omega}) = \underline{H(e^{j\omega})}\, X(e^{j\omega})$$

"convolution" "multiplication"

Suppose $x[n] = A e^{j\omega_0 n} \quad \overset{DTFT}{\longleftrightarrow} \quad X(e^{j\omega}) = 2\pi A \delta(\omega - \omega_0) \qquad |\omega| \le \pi$

$$Y(e^{j\omega}) = 2\pi A H(e^{j\omega}) \delta(\omega - \omega_0) \overset{DTFT}{\longleftrightarrow} y[n] = A |H(e^{j\omega_0})| e^{j(\omega_0 n + \arg\{H(e^{j\omega_0})\})}$$

$|H(e^{j\omega})|$: magnitude response $\quad \arg\{H(e^{j\omega})\}$: phase response

Difference equations

$$x[n-k] \xleftrightarrow{\text{DTFT}} e^{-jk\omega} X(e^{j\omega})$$

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \xleftrightarrow{\text{DTFT}} Y(e^{j\omega}) \sum_{k=0}^{N} a_k e^{-jk\omega} = \sum_{k=0}^{M} b_k e^{-jk\omega} X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}} = \frac{\sum_{k=0}^{M} b_k (e^{-j\omega})^k}{\sum_{k=0}^{N} a_k (e^{-j\omega})^k}$$

Frequency response is a ratio of polynomials in $e^{-j\omega}$

Solving Difference Equations: steady-state response



$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$ transfer function

Frequency Response

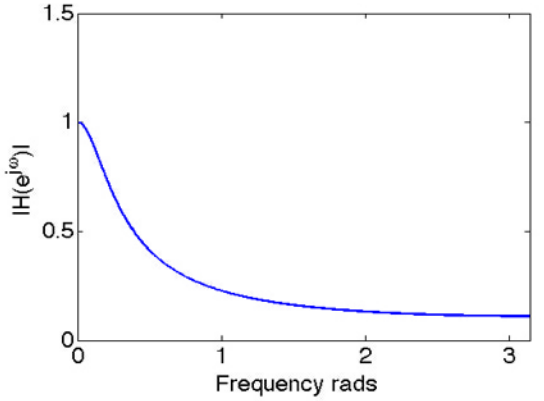
$$H(e^{j\omega}) = H(z)\Big|_{z=e^{j\omega}}$$

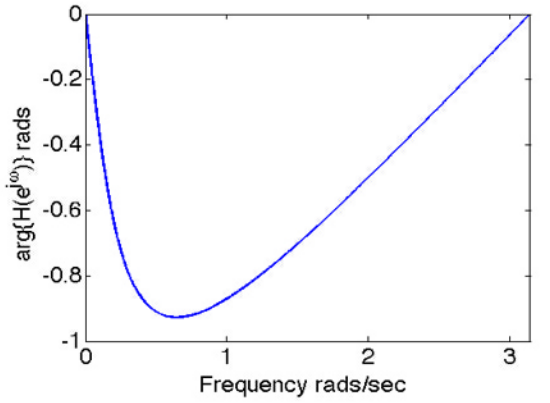
Example: $y[n] - 0.8\,y[n-1] = 0.2\,x[n]$

$N=1, M=0, b_0=0.2$

$a_0=1, a_1=-0.8$

$$H(e^{j\omega}) = \frac{0.2}{1-0.8e^{-j\omega}}$$





$H(e^{j\omega}) \xleftrightarrow{DTFT} h[n] = (0.8)^n u[n]$

- Can compute impulse response using the difference equation
- Find analytical expression for $h[n]$ using $H(e^{j\omega})$

$H(e^{j\omega}) \xleftrightarrow{DTFT} h[n]$

$H(e^{j\omega})$ is a ratio of polynomials in $e^{-j\omega}$ → partial fraction expansion

Inverse DTFT of $H(e^{j\omega})$

1) Root characteristic equation: $1 + a_1 z^{-1} + a_2 z^{-2} + \cdots a_N z^{-N} = 0 \quad (a_0 = 1)$

$$\prod_{\ell=1}^{N} (1 - d_\ell z^{-1}) = 0$$

2) Assuming distinct (non repeated) d_ℓ and $M < N$, expand

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\prod_{\ell=1}^{N} (1 - d_\ell e^{-j\omega})} = \frac{A_1}{1 - d_1 e^{-j\omega}} + \frac{A_2}{1 - d_2 e^{-j\omega}} + \cdots + \frac{A_N}{1 - d_1 e^{-j\omega}}$$

3) Invert each term using $\frac{A}{1 - d e^{-j\omega}} \overset{\text{DTFT}}{\longleftrightarrow} A d^n u[n]$

$$h[n] = A_1 d_1^n u[n] + A_2 d_2^n u[n] + \cdots + A_N d_N^n u[n]$$

Requires $|d| < 1 \iff$ stable system

Example: $y[n] + \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = 3x[n] + 4/3\,x[n-1]$

Find $h[n]$.

$$H(e^{j\omega}) = \frac{3 + 4/3\,e^{-j\omega}}{1 + 5/6\,e^{-j\omega} + 1/6\,e^{-j2\omega}} \xleftrightarrow{\text{DTFT}} h[n]$$

1) Characteristic eqn $1 + 5/6\,z^{-1} + 1/6\,z^{-2} = 0$ or $z^2 + \frac{5}{6}z + \frac{1}{6} = 0$

$$(z + 1/3)(z + 1/2) = 0$$

$$(1 + 1/3\,z^{-1})(1 + 1/2\,z^{-1}) = 0$$

$d_1 = -1/3 \quad d_2 = -1/2$

2) Expand

$$\frac{3 + 4/3\,e^{-j\omega}}{(1 + 1/3\,e^{-j\omega})(1 + 1/2\,e^{-j\omega})} = \frac{A_1}{1 + 1/3\,e^{-j\omega}} + \frac{A_2}{1 + 1/2\,e^{-j\omega}} \Rightarrow \begin{array}{l} A_1 + A_2 = 3 \\ 1/2\,A_1 + 1/3\,A_2 = 4/3 \end{array}$$

$$= \frac{2}{1 + 1/3\,e^{-j\omega}} + \frac{1}{1 + 1/2\,e^{-j\omega}}$$

3) Invert

$$h[n] = 2(-1/3)^n u[n] + (-1/2)^n u[n]$$